UNIT-1

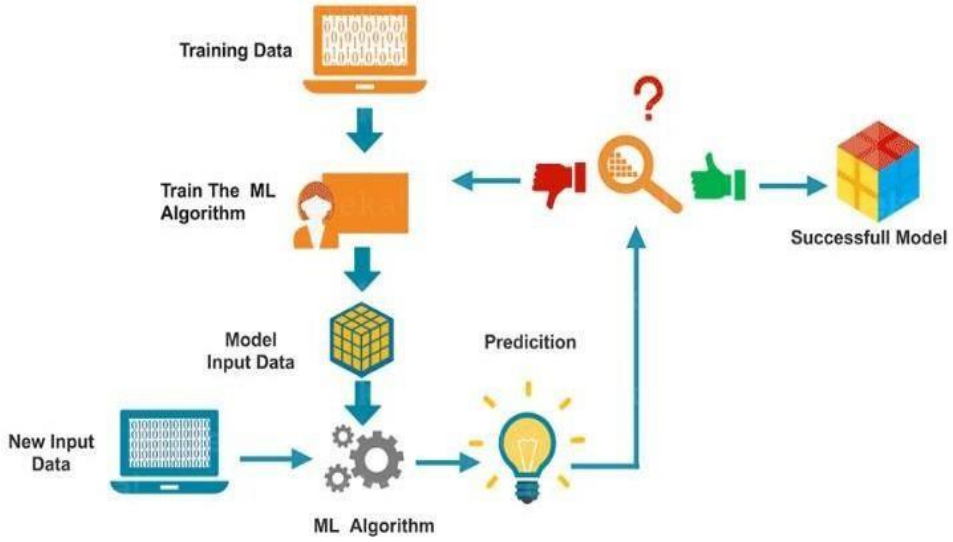
Introduction: Introduction to Machine learning , Supervised learning, Unsupervised learning, Reinforcementlearning. Deep learning.
Feature Selection: Filter, Wrapper , Embedded methods.
Feature Normalization:- min-max normalization, z-score normalization, and constant factor normalization
Introduction to Dimensionality Reduction : Principal Component Analysis(PCA), Linear Discriminant Analysis(LDA)

What is Machine Learning?

Machine Learning is a concept which allows the machine to learn from examples and experience, and that too without being explicitly programmed. So instead of you writing the code, what you do is you feed data to the generic algorithm, and the algorithm/ machine builds the logic based on the given data.

How does Machine Learning Work?

Machine Learning algorithm is trained using a training data set to create a model. When new input data is introduced to the ML algorithm, it makes a prediction on the basis of the model.The prediction is evaluated for accuracy and if the accuracy is acceptable, the Machine Learning algorithm is deployed. If the accuracy is not acceptable, the Machine Learning algorithm is trained again and again with an augmented raining data set



Types of Machine Learning

Machine learning is sub-categorized to three types:

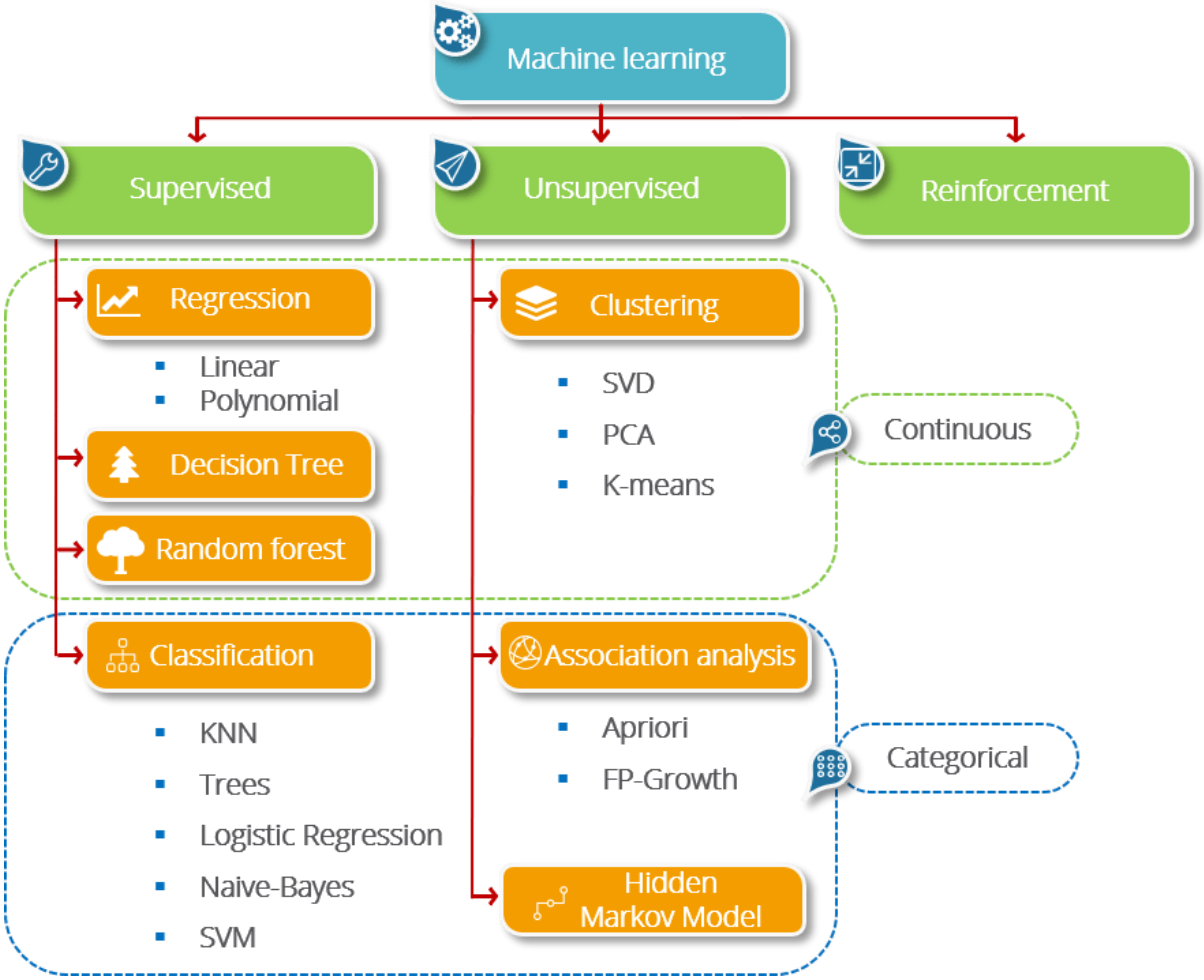
- **Supervised Learning – Train Me!**
- **Unsupervised Learning – I am self sufficient in learning**
- **Reinforcement Learning – My life My rules! (Hit & Trial)**

Supervised Learning is the one, where you can consider the learning is guided by a teacher. We have a dataset which acts as a teacher and its role is to train the model or the machine. Once the model gets trained it can start making a prediction or decision when new data is given to it.

What is Unsupervised Learning?

The model learns through observation and finds structures in the data. Once the model is given a dataset, it automatically finds patterns and relationships in the dataset by creating clusters in it. What it cannot do is add labels to the cluster, like it cannot say this a group of apples or mangoes, but it will separate all the apples from mangoes.
Suppose we presented images of apples, bananas and mangoes to the model, so what it does, based on some patternsand relationships it creates clusters and divides the dataset into those clusters. Now if a new data is fed to the model, it adds it to one of the created clusters.

What is Reinforcement Learning?
It is the ability of an agent to interact with the environment and find out what is the best outcome. It follows the concept of hit and trial method. The agent is rewarded or penalized with a point for a correct or a wrong answer, andon the basis of the positive reward points gained the model trains itself. And again once trained it gets ready to predict the new data presented to it.



Classification of Machine Learning Algorithms Machine Learning algorithms can be classified into:

1. Supervised Algorithms – Linear Regression, Logistic Regression, Support Vector Machine (SVM), DecisionTrees, Random Forest
2. Unsupervised Algorithms – K Means Clustering.
3. Reinforcement Algorithm

1. Supervised Machine Learning Algorithms

In this type of algorithm, the data set on which the machine is trained consists of labelled data or simply said, consists both the input parameters as well as the required output. For example, classifying whether a person is a male or a female. Here male and female will be our labels and our training dataset will already be classified into the

given labels based on certain parameters through which the machine will learn these features and patterns andclassify some new input data based on the learning from this training data.

Supervised Learning Algorithms can be broadly divided into two types of algorithms, Classification and Regression.Classification Algorithms

Just as the name suggests, these algorithms are used to classify data into predefined classes or labels. Regression Algorithms

These algorithms are used to determine the mathematical relationship between two or more variables and the level of dependency between variables. These can be used for predicting an output based on the interdependency of two or more variables. For example, an increase in the price of a product will decrease its consumption, which means, in this case, the amount of consumption will depend on the price of the product. Here, the amount of consumption will be called as the dependent variable and price of the product will be called the independent variable. The level of dependency of the amount of consumption on the price of a product will help us predict the future value of the amount of consumption based on the change in prices of the product.

We have two types of regression algorithms: Linear Regression and Logistic Regression

(a) Linear Regression

Linear regression is used with continuously valued variables, like the previous example in which the price of the product and amount of consumption are continuous variables, which means that they can have an infinite number of possible values. Linear regression can also be represented as a graph known as scatter plot, where all the data points of the dependent and independent variables are plotted and a straight line is drawn through them such that the maximum number of points will lie on the line or at a smaller distance from the line. This line – also called the regression line, will then help us determine the relationship between the dependent and independent variables along with which the linear regression equation is formed.

(b) Logistic Regression

The difference between linear and logistic regression is that logistic regression is used with categorical dependent variables (eg: Yes/No, Male/Female, Sunny/Rainy/Cloudy, Red/Blue etc.), unlike the continuous valued variables used in linear regression. Logistic regression helps determine the probability of a certain variable to be in a certain group like whether it is night or day, or whether the colour is red or

blue etc. The graph of logistic regression consists of a non-linear sigmoid function which demonstrates the probabilities of the variables.

2. Unsupervised Machine Learning Algorithms

Unlike supervised learning algorithms, where we deal with labelled data for training, the training data will be unlabelled for Unsupervised Machine Learning Algorithms. The clustering of data into a specific group will be doneon the basis of the similarities between the variables. Some of the unsupervised machine learning algorithms are K- means clustering, neural networks.

Another machine learning concept which is extensively used in the field is Neural Networks..

3. Reinforcement Machine Learning Algorithms

Reinforcement Learning is a type of Machine Learning in which the machine is required to determine the ideal behaviour within a specific context, in order to maximize its rewards. It works on the rewards and punishment principle which means that for any decision which a machine takes, it will be either be rewarded or punished due to which it will understand whether or not the decision was correct. This is how the machine will learn to take the correct decisions to maximize the reward in the long run.

For reinforcement algorithm, a machine can be adjusted and programmed to focus more on either the long-term rewards or the short-term rewards. When the machine is in a particular state and has to be the action for the next state in order to achieve the reward, this process is called the Markov Decision Process.

What is Normalization in Machine Learning?

Normalization is a scaling technique in Machine Learning applied during data preparation to change the values of numeric columns in the dataset to use a common scale. It is not necessary for all datasets in a model. It is required only when features of machine learning models have different ranges.

Although **Normalization** is no mandate for all datasets available in machine learning, it is used whenever the attributes of the dataset have different ranges. It helps to enhance the performance and reliability of a machine learning model. In this article, we will discuss in brief various Normalization techniques in machine learning, why it is used, examples of normalization in an ML model, and much more. So, let's start with the definition of Normalization in Machine Learning.

Mathematically, we can calculate normalization with the below formula:

Xn = (X - Xminimum) / (Xmaximum - Xminimum)

- Xn = Value of Normalization
- Xmaximum = Maximum value of a feature
- Xminimum = Minimum value of a feature

Example: Let's assume we have a model dataset having maximum and minimum values of feature as mentioned above. To normalize the machine learning model, values are shifted and rescaled so their range can vary between 0 and 1. This technique is also known as **Min-Max scaling**. In this scaling

technique, we will change the feature values as follows:

Case1- If the value of X is minimum, the value of Numerator will be 0; hence Normalization will also be 0.
Xn = (X - Xminimum) / (Xmaximum - Xminimum)

Put X =Xminimum in above formula, we get;

Xn = Xminimum- Xminimum/ (Xmaximum - Xminimum)Xn = 0
Case2- If the value of X is maximum, then the value of the numerator is equal to the denominator; henceNormalization will be 1.

Xn = (X - Xminimum) / (Xmaximum - Xminimum)Put X =Xmaximum in above formula, we get;
Xn = Xmaximum - Xminimum/ (Xmaximum - Xminimum)

What is Dimensionality Reduction?
In machine learning classification problems, there are often too many factors on the basis of which the final classification is done. These factors are basically variables called features. The higher the number of features, the harder it gets to visualize the training set and then work on it. Sometimes, most of these features are correlated, and hence redundant. This is where dimensionality reduction algorithms come into play. Dimensionality reduction is the process of reducing the number of random variables under consideration, by obtaining a set of principal variables. It can be divided into feature selection and feature extraction.

Why is Dimensionality Reduction important in Machine Learning and Predictive Modeling?
An intuitive example of dimensionality reduction can be discussed through a simple e-mail classification problem, where we need to classify whether the e-mail is spam or not. This can involve a large number of features, such as whether or not the e-mail has a generic title, the content of the e-mail, whether the e-mail uses a template, etc. However, some of these features may overlap. In another condition, a classification problem that relies on both humidity and rainfall can be collapsed into just one underlying feature, since both of the aforementioned are correlated to a high degree. Hence, we can reduce the number of features in such problems. A 3-D classification problem can be hard to visualize, whereas a 2-D one can be mapped to a simple 2 dimensional space, and a 1-D problem to a simple line. The below figure illustrates this concept, where a 3-D feature space is split into two 1-D feature spaces, and later, if found to be correlated, the number of features can be reduced even further

Components of Dimensionality Reduction
There are two components of dimensionality reduction:

- **Feature selection:** In this, we try to find a subset of the original set of variables, or features, to get a smallersubset which can be used to model the problem. It usually involves three ways:

1. Filter
2. Wrapper
3. Embedded

- **Feature extraction:** This reduces the data in a high dimensional space to a lower dimension space, i.e. a spacewith lesser no. of dimensions.

Methods of Dimensionality Reduction